F4 second pass: streaming signed stitching and anchored sections

26 October 2026 · TASK-20261026-b71abe · Phase A source preparation.

No second-pass panel, pilot, benchmark or arithmetic program has run. The prospective source, draft protocol, frozen-analysis candidate and exporter are prepared for root inspection, source pinning and protocol freeze. Data collection still requires the explicit root release and a successful real S3 checkpoint gate. This report contains derived mathematics and planned checks, not observed second-pass results.

What changes from the first pass

The first pass used a dense signed-incidence kernel as a correctness oracle and deliberately engineered coincident outside points. The new source adds a streaming weighted two-endpoint reducer with arithmetic witness DAGs. It retains signs, multiplicities, one-endpoint columns, and closures of unbalanced components. A lone unbalanced triangle must not be called a cancellation. Separate polynomial, Jacobian-point and dense-kernel routines check the producer's output; they are separate implementations in this producer session, not an independent scientific review.

The fixed observation panel specifies 128 exact pole-three sections: sixteen hash-derived slopes through each of eight base anchors G through 8G over the actual P-256 field. The inverse-closed base has sixteen inverse orbits, represented by G through 16G. The program retains every factorization, nonsplit residual, rejection, complete relation, graph input and failure. No residual zero is planted in the natural panel. Public scalar-known anchors and the constructed graph controls are stated explicitly.

Derived before-data facts

For nonzero P in the published prime-order group, anchored pole-three sections normalize to y-y_P+t(x-x_P), one for every t in F_p. Residual unordered rational pairs are the orbits of Q -> -P-Q on E(F_p) minus {O,-P}. There is one fixed point. Thus exactly (r-1)/2 slopes are fully rational; the ideal uniform-slope probability is (r-1)/(2p), including repeated zeros. It is not a factor-base or collision probability. Fixed hash-derived input fractions are descriptive only.

For one anchor, inverse-orbit columns form a path [k]--[k+1] plus a terminal column of weight two. Removing the base anchor gives a boundary one-outside column from the tangent at P. This is an incidence/half-edge description; an ordinary loop display has a different valence convention. Combining eight anchors does not license a random-graph assumption.

The reducer stores pivots with at most two outside entries. Every zero residual emits a witness with a unique latest input coefficient one. The resulting independent witnesses number L-rank(A), so they span the full outside kernel. Their base-row image rank is rank(D)-rank(A); it can be smaller or zero. Exact integer representatives cancel modulo r, with any integer-lift multiples of r retained. Details and exceptional cases are in the prospective derivation (theory.md).

Fixed controls and accounting

Four export cells each represent an entire arm-panel, numeric instance 0, replicate 0. The natural arm generates the 128 unique functions. Its A/A arm reuses those exact cases in reverse graph order and checks structural rank equality; it does not create another 128 samples. Seven positive families cover balanced cycles of lengths 4, 8 and 16, paired unbalanced cycles, and three explicitly abstract weighted/zero-outside matrix controls. Four negative mechanisms cover a lone unbalanced triangle and corrupted cancellation, sign and norm-factor certificates. The omitted-factor control uses a pinned historical nonsplit input, clearly separated from new data.

Source checks distinguish the pinned implementation commit from runtime HEAD, verify selected Git-object bytes and the actual Python executable, and retain host, commands, diagnostic rusage and exact object/call counts. All clocks have wall_valid=false. Scalar-call and factorization counts, DAG storage, verifier work and witness expansion are not a complete GAE, attack cost or matched performance comparison. No official scientific or model-policy admission is claimed.

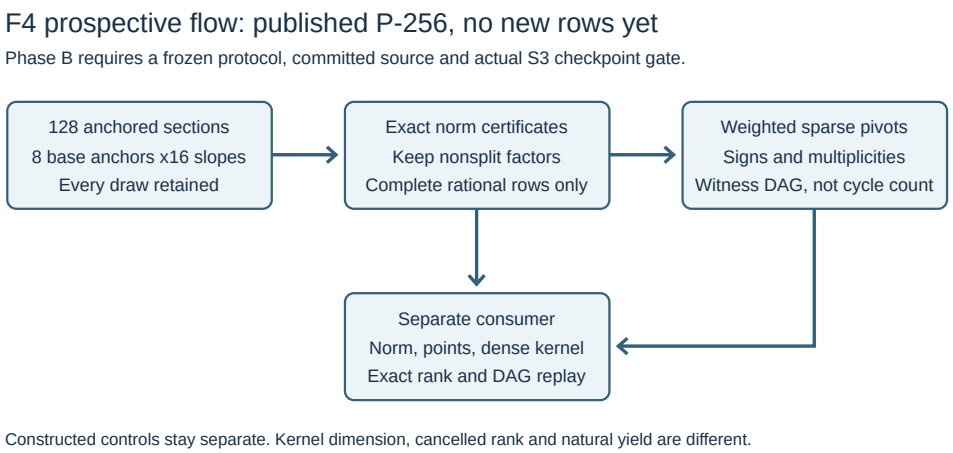

Figure 1: Prospective F4 execution and certificate flow

The source review and transfer limits are in literature.md (literature.md). The genus-at-least-two double-large-prime complexity theorem is not transferred to P-256. This pass leaves natural large-base yield, target attachment, full solve/rank costs and amortized behavior open. A bounded absence of collisions would not prove an impossibility theorem.

After an authorized execution, the fresh run directory contains its own immutable result.json, all raw certificates, rows.jsonl, analysis graph and report, and unified export.json. This preparation report will not be silently replaced by a result report.